

Collaborative Filtering for Movie Recommendations



SVD vs Item-Based Collaborative Filtering

Netflix Prize Dataset · Evaluation Framework · 8 Performance Metrics

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~98%

Global matrix sparsity

480K+

Unique users rated

8

Evaluation metrics

2

Model paradigms

The Recommendation Challenge



Selection Bias

Users rate content they expect to enjoy. Missing entries $\neq$ negative feedback. Naive models treating zeros as dislikes are fundamentally flawed.



Popularity Concentration

A tiny cluster of blockbusters accumulates the vast majority of interactions. Models optimise for these dense areas, ignoring 65%+ of the catalog.



Extreme Sparsity (98%)

The full user–item interaction matrix is $\sim 98\%$ empty. Memory-based lookups fail when items share almost no co-raters.

Dataset at a Glance

480K

Active users

17K+

Unique movies

8M+

Total ratings

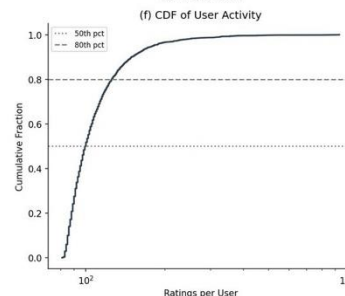
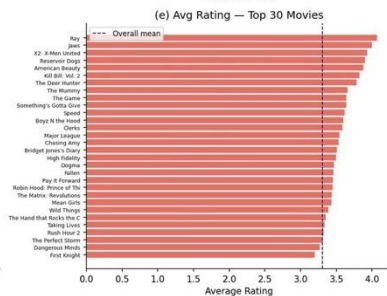
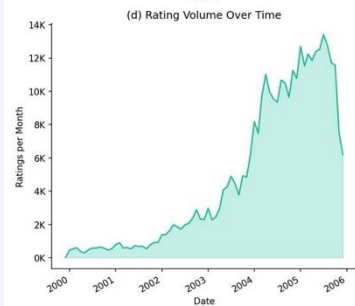
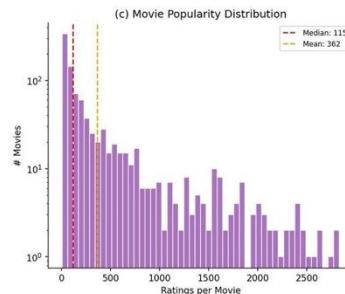
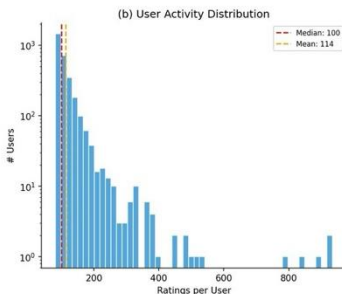
3.3–3.5

Avg rating range

Goal: Design, implement, and evaluate recommendation models that balance prediction accuracy with catalog coverage.

(a) Rating Distribution

Rating	Count (%)
1	8.6%
2	13.4%
3	33.1%
4	28.9%
5	16.0%



Modal score = 3★ (33.1%). Only 8.6% are 1-star — classic selection bias.

Movie median = 115 ratings; mean = 362.
Blockbusters skew totals.

Rating volume exploded 2003–2005. Monthly avg stayed stable at 3.3–3.5.

Even top 80×80 submatrix is 19.6% missing. Full matrix $\approx 98\%$ sparse.

Two Collaborative Filtering Paradigms

Memory-Based · Item-CF

Item-Based Collaborative Filtering

Finds items with similar rating patterns across users. Uses Adjusted Cosine Similarity to correct for individual rating scale biases.

$$Sim(i,j) = \frac{\sum (R_{ui} - \bar{R}_u)(R_{uj} - \bar{R}_u)}{\sqrt{\sum (R_{ui} - \bar{R}_u)^2 \cdot \sum (R_{uj} - \bar{R}_u)^2}}$$

- ✓ Interpretable (neighborhood-based)
- ✓ No training phase needed
- ✗ Degrades with very sparse overlaps
- ✗ Lower ranking precision

Model-Based · Latent Factor

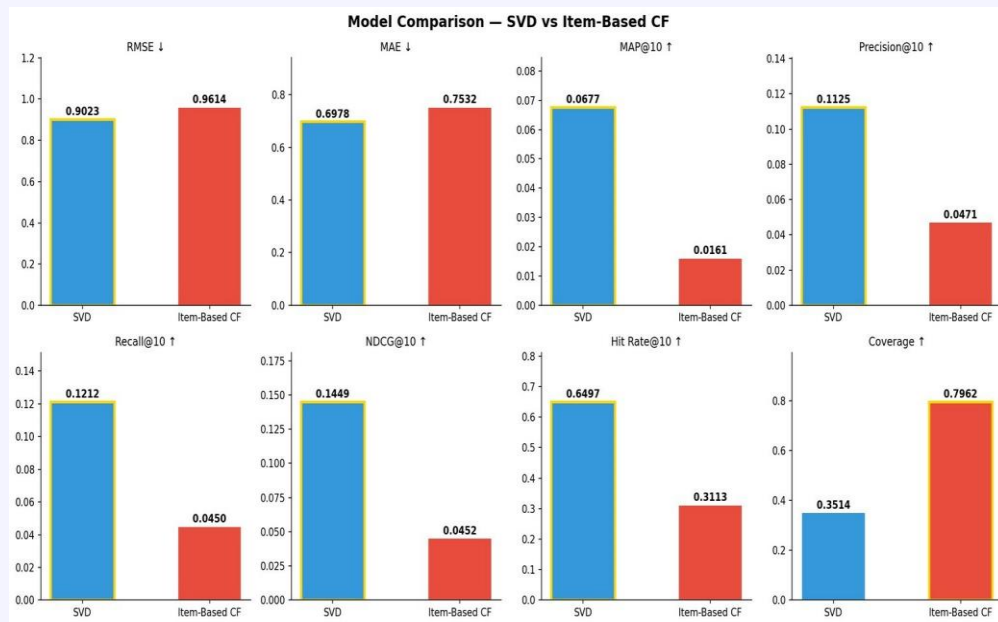
Singular Value Decomposition (SVD)

Maps users and items into a shared latent space. Learns global biases + latent preference/characteristic vectors via regularised SGD on known ratings.

$$R_{ui}^{\#} = \mu + b_u + b_i + p_u^T q_i$$

- ✓ Handles 98% sparsity via dimensionality reduction
- ✓ Captures cross-genre latent preferences
- ✗ Higher training cost than memory-based
- ✗ Tends toward popularity concentration (35% cov.)

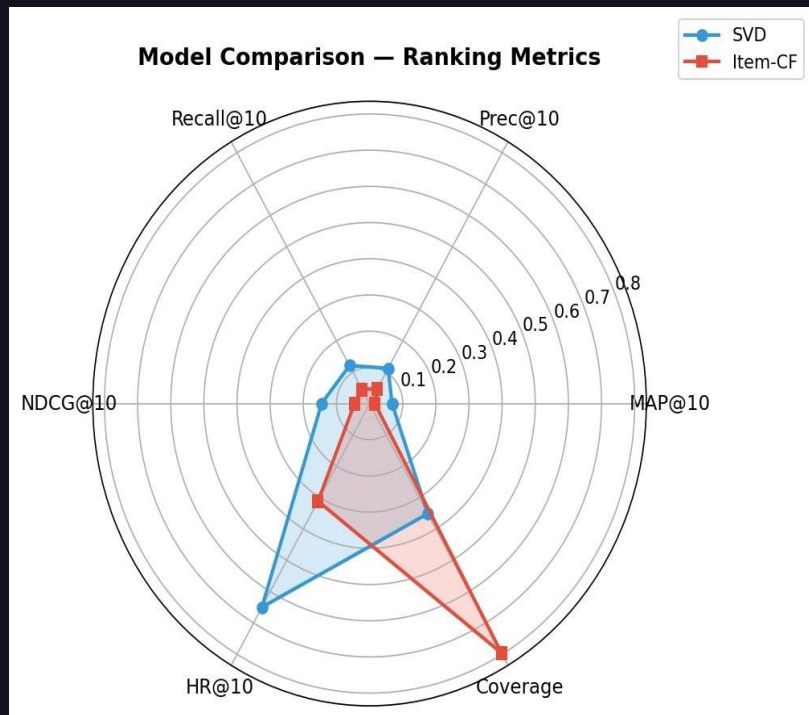
Model Performance — K = 10 Recommendations



Metric	SVD	Item-CF
RMSE ↓	0.9023 ✓	0.9614
MAE ↓	0.6978 ✓	0.7532
MAP@10 ↑	0.0677 ✓	0.0161
Prec@10 ↑	0.1125 ✓	0.0471
Recall@10 ↑	0.1212 ✓	0.0450
NDCG@10 ↑	0.1449 ✓	0.0452
HitRate@10 ↑	0.6497 ✓	0.3113
Coverage ↑	0.3514	0.7962 ✓

SVD: 4.2× better MAP@10 | Item-CF: 2.3× broader coverage

Coverage vs. Accuracy — The Inescapable Dilemma



SVD — Precision Champion

Wins every ranking metric. $\text{MAP@10} = 0.0677$ ($4.2\times$ Item-CF). Latent biases handle sparsity elegantly by decoupling user leniency from taste.

Item-CF — Diversity Champion

Item-Item Cosine Similarity — Top 30 Movies

79.6% catalog coverage vs SVD's 35.1%. Explores item neighborhoods broadly — ideal for serendipity and long-tail asset monetisation.

Similarity Space Insight

Most top-30 movies cluster in 0.4–0.8 cosine similarity range. Lilo & Stitch and North by Northwest show near-zero similarity — distinct audience demographics correctly isolated.

Recommendation Examples — User Profiles

USER 1333 — SVD

1 **Seven Samurai** ★ 4.11

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2 **Christmas with The Simpsons** ★ 3.94

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3 **Animal Crackers** ★ 3.88

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4 **Bill Cosby: Himself** ★ 3.87

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5 **A Night at the Opera** ★ 3.86

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USER 1333 — Item-CF

1 IFilm: Deranged ★ 3.03

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2 Hamlet ★ 2.97

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3 Rebecca: Bonus Material ★ 3.08

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4 The Return ★ 3.06

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5 The Agronomist ★ 3.03

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SVD surfaces critically acclaimed classics with high-confidence predictions (bias-corrected). Item-CF explores long-tail niche titles via local similarity — ratings are lower but catalog diversity is higher.

Key Insights & Recommended Next Steps

Latent > Local for Ranking

SVD's bias decomposition ($\mu + b_u + b_i + p_u^T q_i$) separates global sentiment from taste. Item-CF lacks this, causing similarity scores to degrade in sparse neighborhoods.

Coverage–Accuracy Dilemma

Optimising for RMSE/NDCG concentrates SVD on a 35% safe subset. Item-CF's 79.6% coverage enables serendipity and long-tail content monetisation.

Selection Bias is Structural

The positive rating skew (8.6% ones, 33.1% threes) reflects user pre-screening, not true sentiment. Missing entries must never be treated as negative feedback.

FUTURE IMPROVEMENTS

Hybrid Ensemble

Blend SVD predictions with Item-CF diversity scores using a linear stacking layer to balance accuracy and coverage.

Rich Side-Info

Integrate genre, director, and user demographics via LightFM or Deep Factorisation Machines to resolve cold-start.

Post-Hoc Re-ranking

Apply Maximal Marginal Relevance (MMR) to SVD output — deliberately boost long-tail items and suppress over-exposed blockbusters.